UNIT 4: Linear models

Some statistical tests

- * NOANS
- * NOANS
- * NOANS

Linear models of observations

- * t test
  - * including paired t, one-sample t
- * regression
  - * including Pearson correlation
- * anova
- * ancova

Outline

Approach and assumptions

Examples

Complications

- Multi-parameter variables
- Multiple predictor variables
- Interactions

What is a linear model?

- A *fit* to a **response variable** based on assumptions about its relationship to one or more **predictor variables**
- *Inferences* about parameters based on assumptions about the *conditional* distribution of the response variable
  - Assume the response variable has "true" distribution for any value of the predictor variables
    - * An example
- A linear model gives the response variable as a linear combination of the *parameters* it is estimating
  - * It is not necessarily linear with respect to predictor variables!

Linear model?

y is response, x is predictor

- $y = a + bx$
- $y = a + bx + cx^2 + dx^3$
- $y = \frac{a}{b+x}$
- $y = ax^b$
- $\log(y) = a + b\log(x)$
- * 1, 2 and 5 are lms
  - * Linear in a, b, ...
- * 4 and 5 are the same equation, but imply a different *distribution* for y

Linear model?

- What if x is a categorical variable?
  - * *Any* response to a categorical variable can be encoded as a linear model!

Power

- Linear models are very efficient
  - They find the best "least-squares" fit with simple linear algebra
- They can be very powerful at detecting effects
  - They can also be over-sensitive to outliers
  - *Spiderman: with great power comes great responsibility*



Assumptions

- Linear-model *fits* are basically always correct, but can be hard to interpret
  - You can find the "best" parameters
  - But you need assumptions to know what that means
- Linear-model *inferences* (parameter estimates P-values, CIs) depend on assumptions about *residuals*
- A residual is $y - \hat{y}$: the difference between an observed value y and a model prediction $\hat{y}$
- lm inference assumes that residuals are independent and normally distributed, with constant variance
  - The mean of the residuals is always 0
  - * Otherwise the machine would find a better linear model

Variables

- What kind of variables can we use?
- Response
  - * We assume the response variable is normally distributed (conditional on the predictor variables)
  - * Therefore it must be continuous
- Predictors
  - * The linear model framework gives us a lot of flexibility to construct a model
  - * Predictor variables can be of any type
  - * e.g., Continuous, integer, ordinal, nominal

Fitting in R

- $\mathrm{lm}(y \sim 1 + x + c)$
  - Use y as a response variable, x and c as predictors
  - 1 refers to the *intercept*, which is a constant term
- $\mathrm{lm}(y \sim x + c)$
  - * *Also* has an intercept
  - * intercepts are usually good
  - * $\mathrm{lm}(y \sim 0 + x + c)$ to suppress it
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No predictor

- **One-sample t-test**
  - e.g., what is the mean weight of an African elephant?
- Assumption: population is normally distributed
- Example: fev

Check the assumption

- * Is the sample normally distributed?
  - * No! Nothing in biology is perfect
- * Does it look close to normal?
  - * Should you consider another test, or another question?
    - * Permutation test of the mean
    - * Test the median instead
    - * Test the mean on a transformed scale

Interpreting diagnostics

- Your linear model assumptions are not met exactly
  - You can't use failure to reject the null hypothesis as evidence that your assumptions are OK
- There is no universally accepted way to decide whether you are close enough!
  - When sample size gets larger, your ability to reject the null hypothesis get stronger
  - But concerns often get smaller
    - Particularly normality concerns
    - * central limit theorem!
- We may circle back to this for projects

Interpreting results

- Generally, avoid R's summary table!
  - * It shows P values but not CIs
- Focus on estimates and confidence intervals
- But how to interpret these?
  - * Pay attention to units
  - * You need an outside reference
  - * Statistics is not magic

Binary predictor

- Two-group test
  - Predictor with two values (could be continuous or categorical)
  - Can men hold their breath longer than women?
  - Are mice more active at 10 degrees or 20 degrees C?
  - Generally interest in:
    - * The sign of the difference
    - * Also the magnitude (size)
- Example mouseActivity

Testing assumptions

- * Residuals should look approximately normal
- * and approximately the same across the two groups
- Otherwise? Same answers as above
  - * Permutation test
  - * Different scale
  - * Medians

R model parameters

- Intercept (constant) is the estimated value at:
  - zero for continuous predictors
  - "baseline" for categorical predictors
- Other parameters
  - slopes for continuous predictors
  - differences for categorical predictors

Effect sizes

- What are the units of our parameter in this example?
  - * ml CO2/h for the mouse activity
    - * Or: (ml CO2/h)/degC!
  - * inches for the height . . .
- How should we know if they are large or not?
  - * Compare to population statistics, not inferential statistics
  - * We want to know if the value is large compared to something real, not compared to our own uncertainty!
  - * e.g., total value, variation in population, difference between sexes, difference between species . . .
  - * Use a log scale!

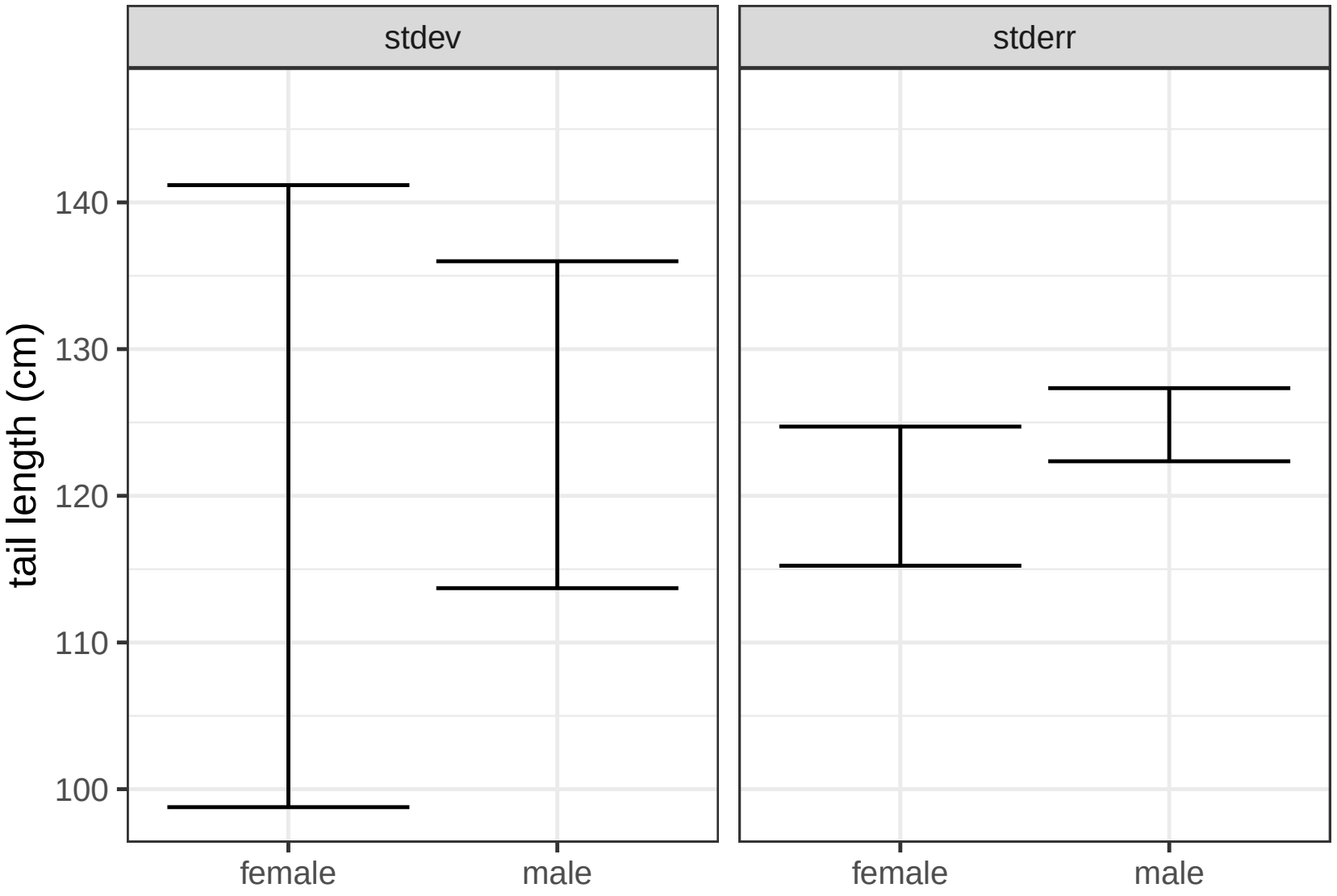
Measures of size

- We can examine size of a parameter in two ways
- Certainty
  - * t value, P value
  - * This is usually our certainty about the sign of something – but what?
- Effect
  - * Unit-ed value, ratio of value to something with same units
  - * or a log ratio

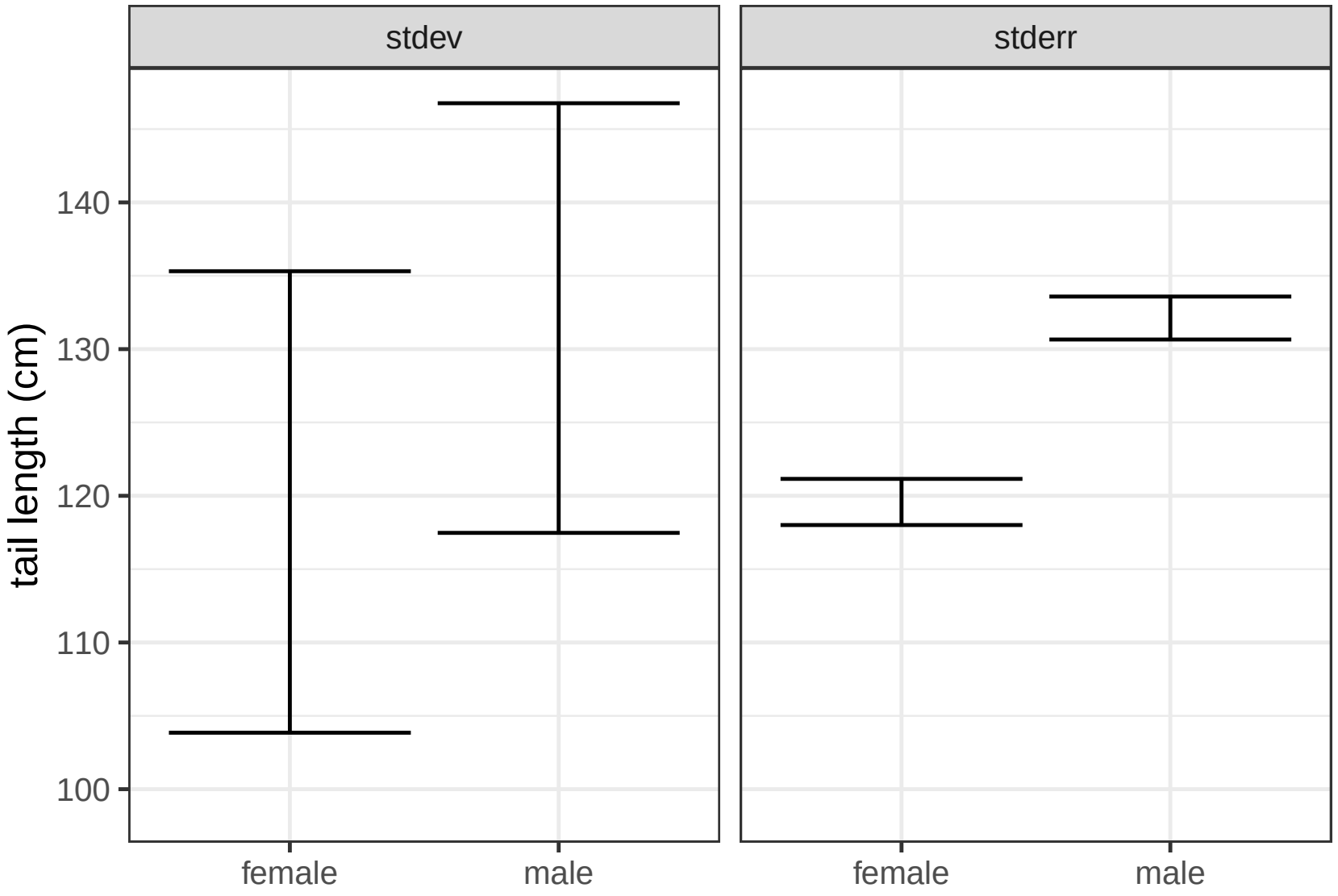
Measures of spread

- Standard deviation
  - * How much variation we see in the population
  - * Compare this to understand the effect estimate
- Standard error (or confidence interval)
  - * How much variation we would expect to see if estimated the parameter again
  - * Compare this to understand the certainty of something

Small sample



Large sample



Parameterizing

- Different two-group models have the same fit
  - * predicted mean for each observation
  - * estimated standard deviation
- but different parameterizations
  - baseline and difference (default for categorical variables)
  - intercept and slope (default for continuous variables)
  - mean and difference
- lm does not work with too many parameters
  - * You would need to choose two in this case
  - * Because we're interested in a mean for each group

Paired t-test

- Example: do people have more joint erosion on their dominant hand?
  - * Can't treat my left hand and right hand as independent!
- How to do it?
  - * The paired t-test is equivalent to a one-sample test on the *differences*!
  - * Procedure is the same, but now we're particularly interested in the *sign* of the estimate
- CLASS See notes for finger-length example

Continuous predictor

- Regression analysis
- Typically focus on the slope
- Pay attention to baseline as well as intercept
- Example: fev.final.R

Diagnostics

Interpretation

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What if you don't think response is a straight line?

- Transform the response
- Transform the predictor
- Fit a curve

Categorical predictor

- Anova test
- We *add* $n-1$ new parameters for a variable with n categories
  - We also add $n-1$ "lm variables" to the calculation
  - These are often called "dummy variables"
  - They code which category each observation is in, using 0s and 1s

Variables and parameters

- To fit a quadratic, we need two parameters for the response
- This also means we need two "lm variables" for the model
  - * x and x^2
- We might want to fit other sorts of curves (typically a spline)

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